

Active Directory Home Lab

Windows Server 2022 Domain Administration Project

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1. Project Overview

I built this project because I wanted to understand Active Directory by actually setting up and managing a Windows domain myself. I created a small lab with Windows Server 2022 and a Windows 11 client and added more features as I worked through the project. This included users and groups, Group Policy, file permissions, PowerShell automation, security auditing, DHCP, drive mapping, and backup.

2. Project Objectives

The main goals of this project were to:

- Build and configure a Windows Server domain.
- Create an organized Active Directory structure for different departments.
- Manage users and access through security groups.
- Join a Windows client computer to the domain.
- Apply security settings through Group Policy.
- Configure department-based file sharing and permissions.
- Automate user onboarding with PowerShell.
- Verify successful and failed logon activity through Event Viewer.
- Practice troubleshooting issues within a virtual lab environment.

3. Lab Environment

Component	Configuration
Hypervisor	Oracle VirtualBox
Domain Controller	DC01
Client Computer	CLIENT-01
Server OS	Windows Server 2022 Evaluation
Client OS	Windows 11
Domain Name	NOLANCYBER.local
DNS Server	DC01
Active Directory	Active Directory Domain Services (AD DS)
PowerShell	Used for user provisioning and automation

I built the lab with two virtual machines in VirtualBox. DC01 runs Windows Server 2022 and acts as the domain controller, while CLIENT-01 is a Windows 11 computer joined to the domain. Using virtual machines let me build and test the whole environment on my own computer.

4. Technologies Used

Operating Systems

- Windows Server 2022 Evaluation
- Windows 11

Virtualization

- Oracle VirtualBox

Windows Server Roles and Features

- Active Directory Domain Services (AD DS)
- Domain Name System (DNS)
- Group Policy Management

Administration Tools

- Active Directory Users and Computers (ADUC)
- Group Policy Management Console (GPMC)
- Event Viewer
- PowerShell

Security & Access Management

- Organizational Units (OUs)
- Security Groups
- NTFS Permissions
- Shared Folder Permissions
- Windows Security Auditing

Automation

- PowerShell Scripting
- CSV user provisioning (Import-Csv)

Networking

- TCP/IP
- Static IP Address Configuration
- Domain Name Resolution (DNS)

5. Network and Domain Configuration

Setting	Configuration
Domain Name	NOLANCYBER.local
Domain Controller	DC01
Client Computer	CLIENT-01
DC01 IP Address	192.168.1.10
CLIENT-01 IP Address	192.168.1.20
DNS Server	192.168.1.10
Network Type	Internal VirtualBox Network

The first step of the project was creating a Windows domain using Windows Server 2022. After installing Active Directory Domain Services and DNS on the server, I promoted the server to a domain controller and created the **NOLANCYBER.local** domain.

Once the domain was running, I configured a Windows 11 virtual machine with a static IP address and pointed its DNS server to the domain controller. After verifying network connectivity, I joined the client computer to the domain and confirmed that domain users could successfully log in.

6. Active Directory Structure

I organized the domain using separate OUs for each department, along with security groups and user accounts. I wanted the structure to be easy to manage as I added more users and started applying permissions and policies

Organizational Units (OUs)

I created a separate OU for each department instead of putting every user in the default Users container. This kept the accounts organized and also gave me more control over where I could apply policies.

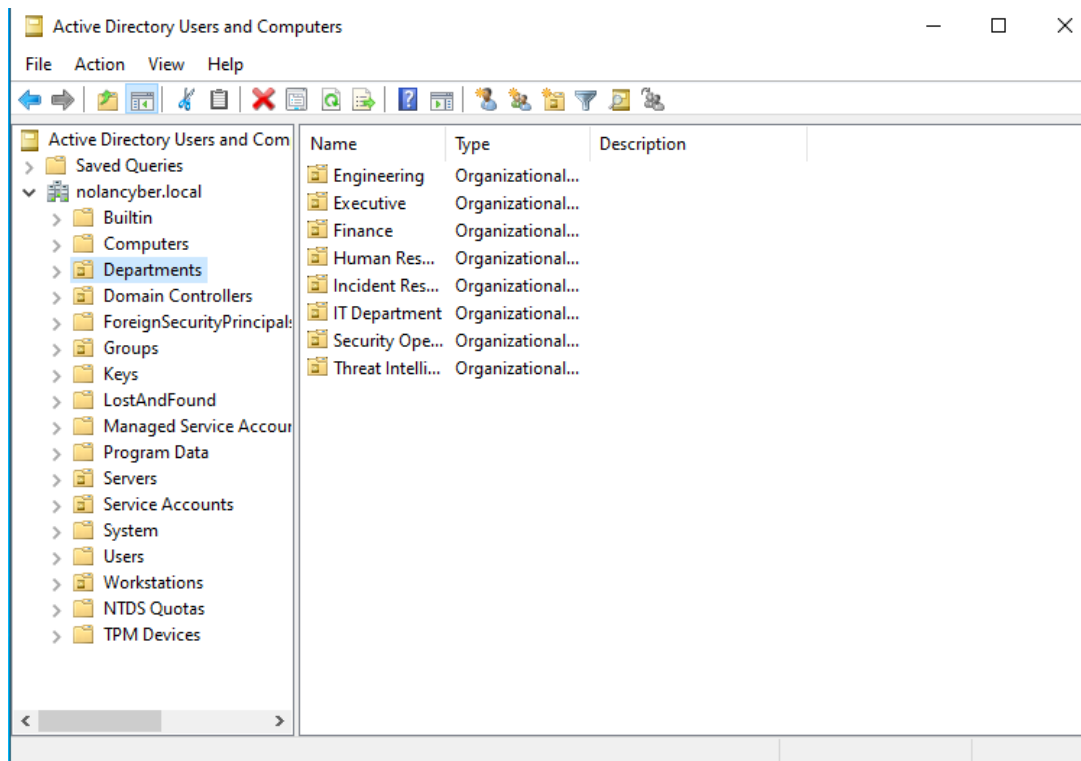


Figure 1. Active Directory Organizational Unit structure used to organize departments and simplify administration.

Security Groups

I created security groups for each department and used those groups when assigning permissions instead of giving access to users one at a time. This meant that when I added a new employee, I could put them in the right group rather than changing the folder permissions for their individual account.

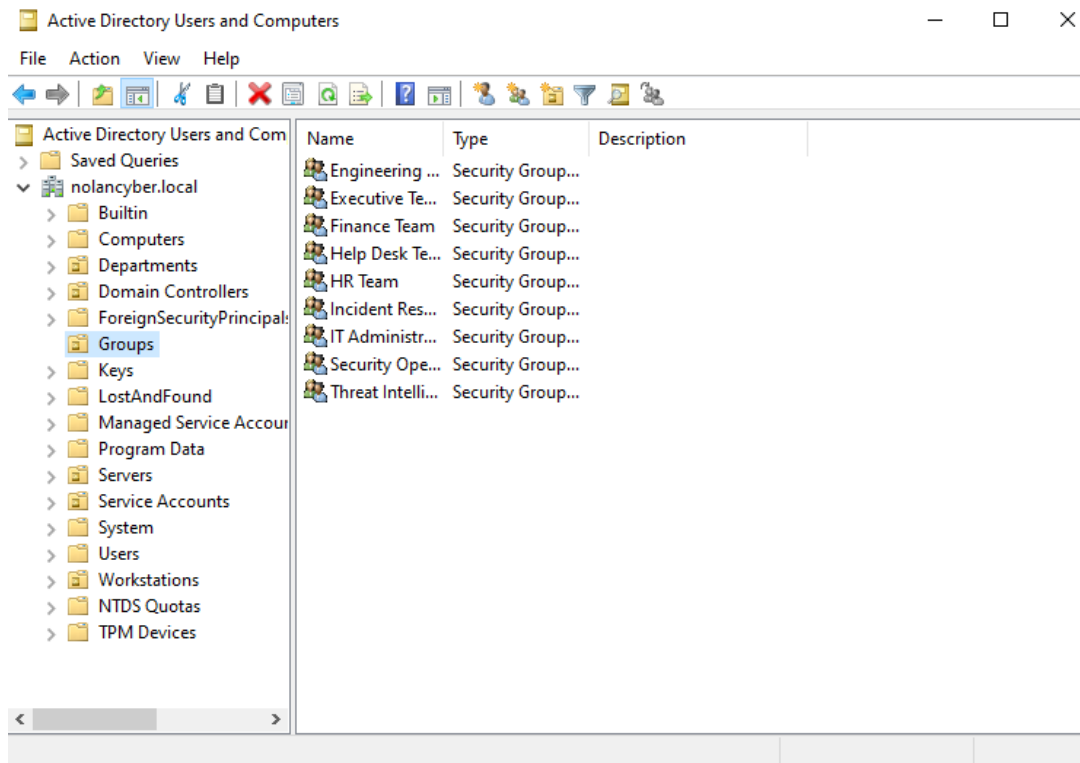


Figure 2. Department-based security groups created to simplify permission management and support role-based access control.

User Accounts

User accounts were created for each department to simulate employees in a small business environment. Each account was placed in the appropriate Organizational Unit and assigned to the correct security group based on its role. This organization made it easier to manage permissions and test access to shared folders throughout the project.

To keep the environment consistent, I followed a simple username naming convention using the employee's first initial and last name (for example, **mcurie** and **alennox**). Later in the project, I also automated user creation with PowerShell by importing employee information from a CSV file, allowing multiple accounts to be created and assigned to the correct Organizational Units and security groups.

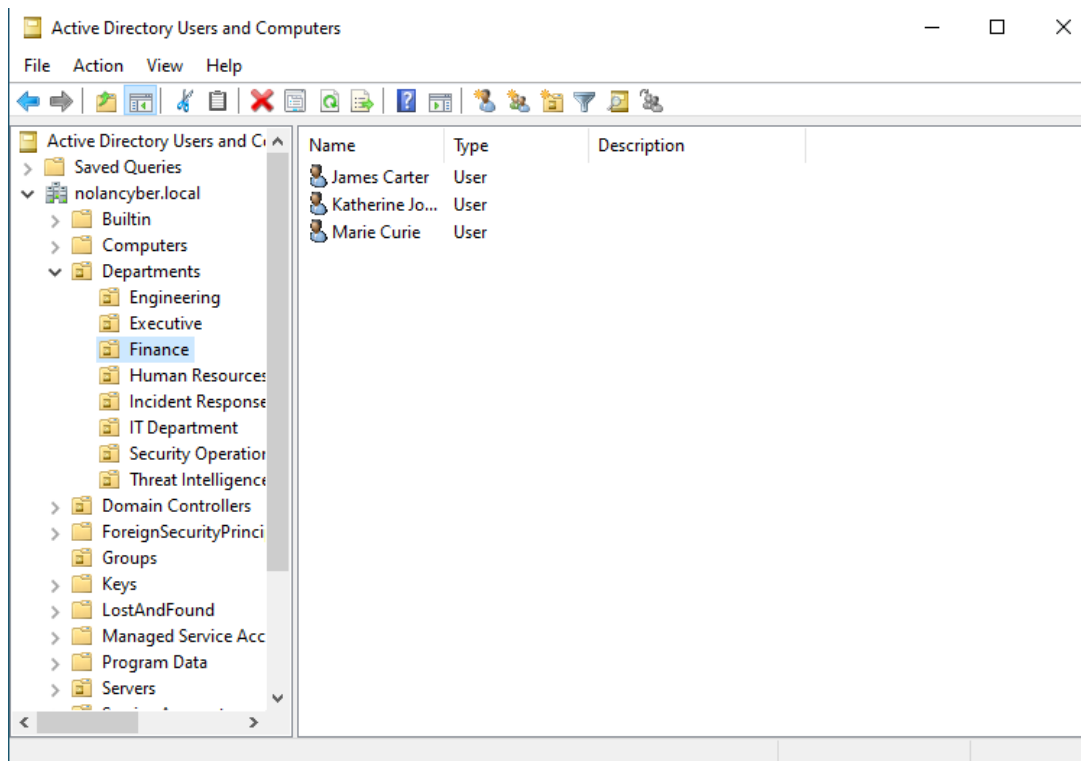


Figure 3. User accounts organized by department using a consistent naming convention and group membership to support centralized administration.

7. Group Policy Configuration

I used Group Policy to manage settings across the domain instead of configuring them separately for each user or computer. I started with password and account policies and later added other policies as I built out the lab.

Password and Account Policies

I configured password complexity, password history, maximum password age, and account lockout settings. This let me practice managing account security from one place and see how the settings affected domain users.

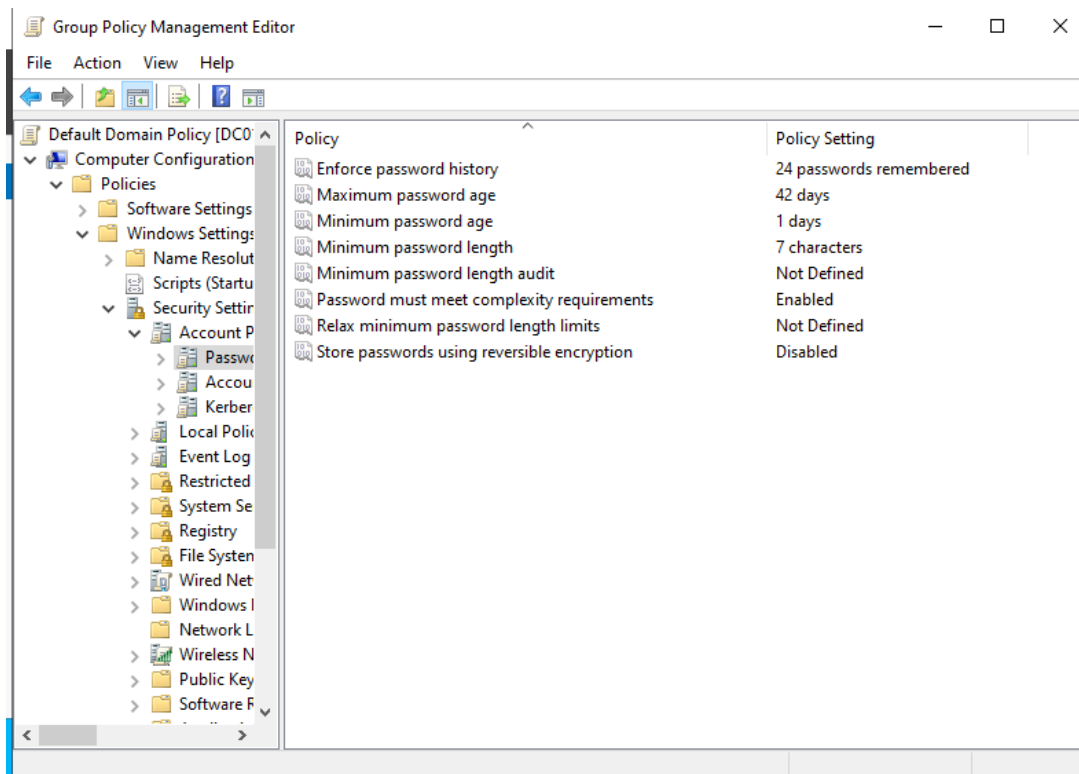


Figure 4. Password and account policies configured through Group Policy to improve domain security.

Security Baseline

I also created a security baseline GPO for the Windows 11 workstation. This gave me a place to apply additional security settings to domain computers beyond the default configuration.

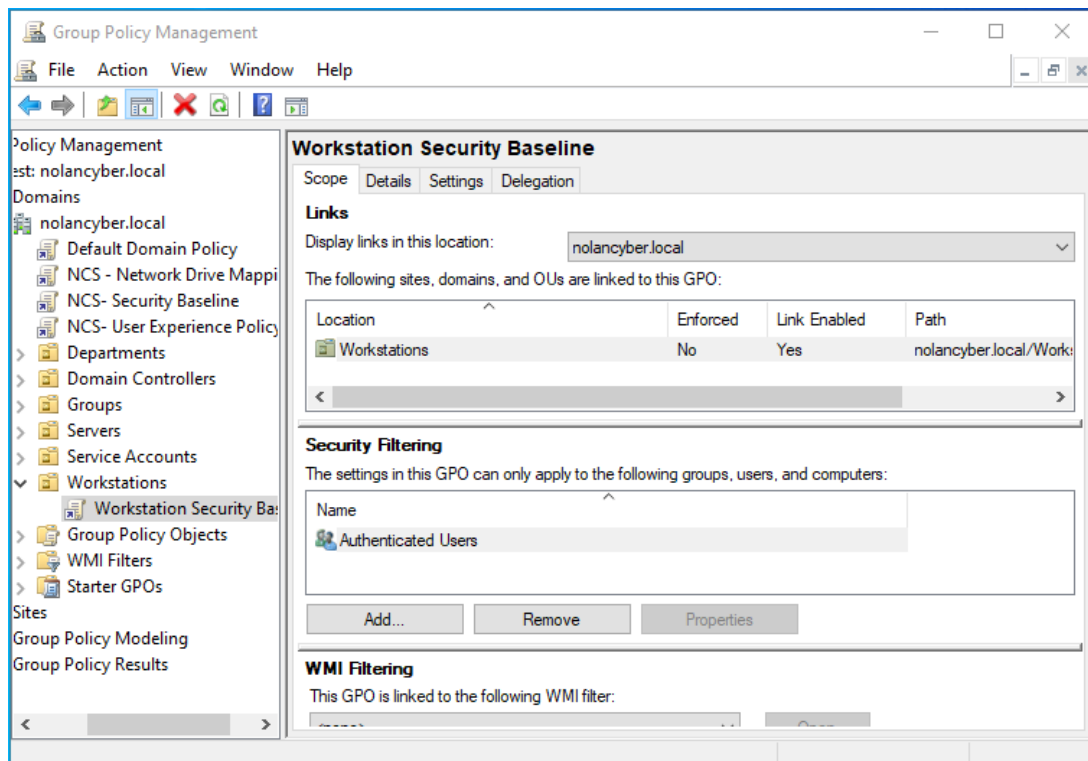


Figure 5. Security baseline Group Policy Object applied to strengthen workstation security.

Interactive Logon Banner

An interactive logon banner was configured to display a legal notice before users signed in. This type of policy is commonly used in organizations to notify users that systems are intended for authorized use and that activity may be monitored.

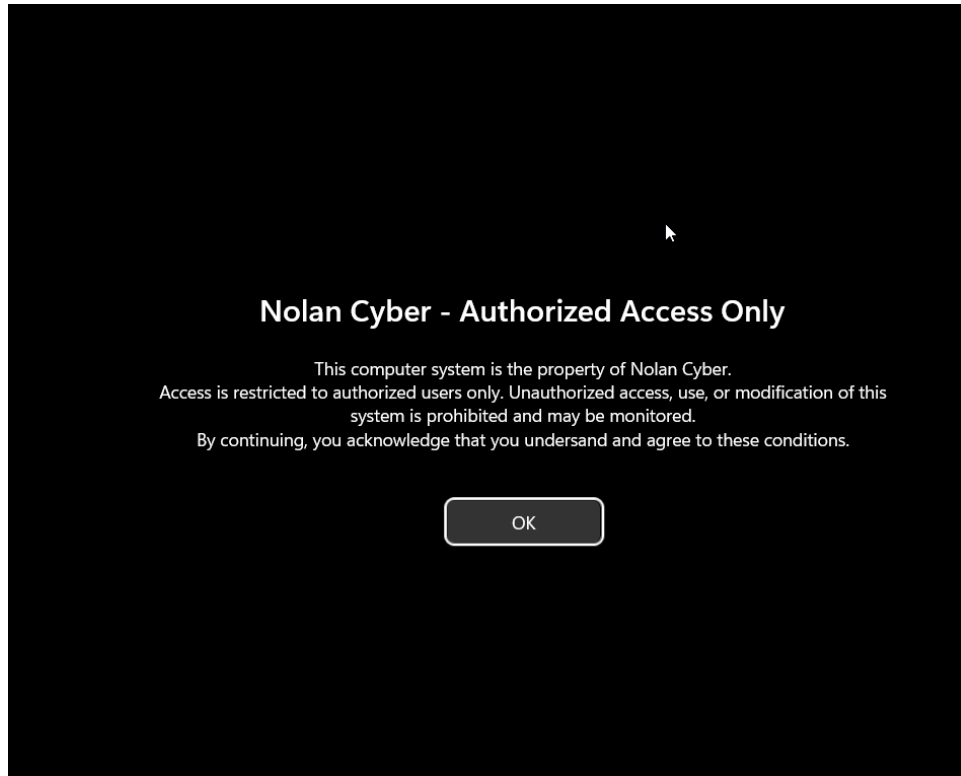


Figure 6. Interactive logon banner displayed before user authentication.

8. File Sharing and Permissions

I created shared folders for different departments and used both share and NTFS permissions to control who could access them. I assigned access through security groups instead of individual user accounts so permissions were easier to manage.

Shared Folder Configuration

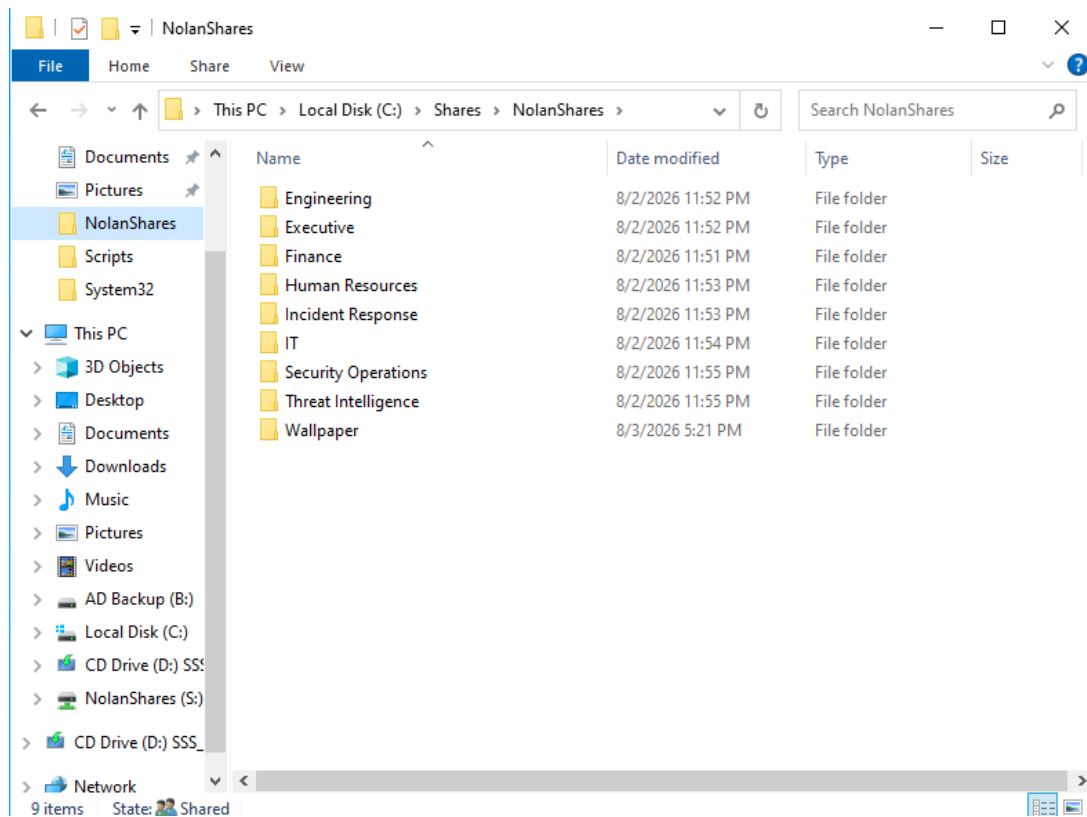


Figure 7. Department folders created on the shared file server for role-based access.

Permission Testing

After configuring permissions, I tested access by signing in as users from different departments. Each user was able to access their assigned department folder while being restricted from folders belonging to other departments. This confirmed that the security groups and folder permissions were working as intended.

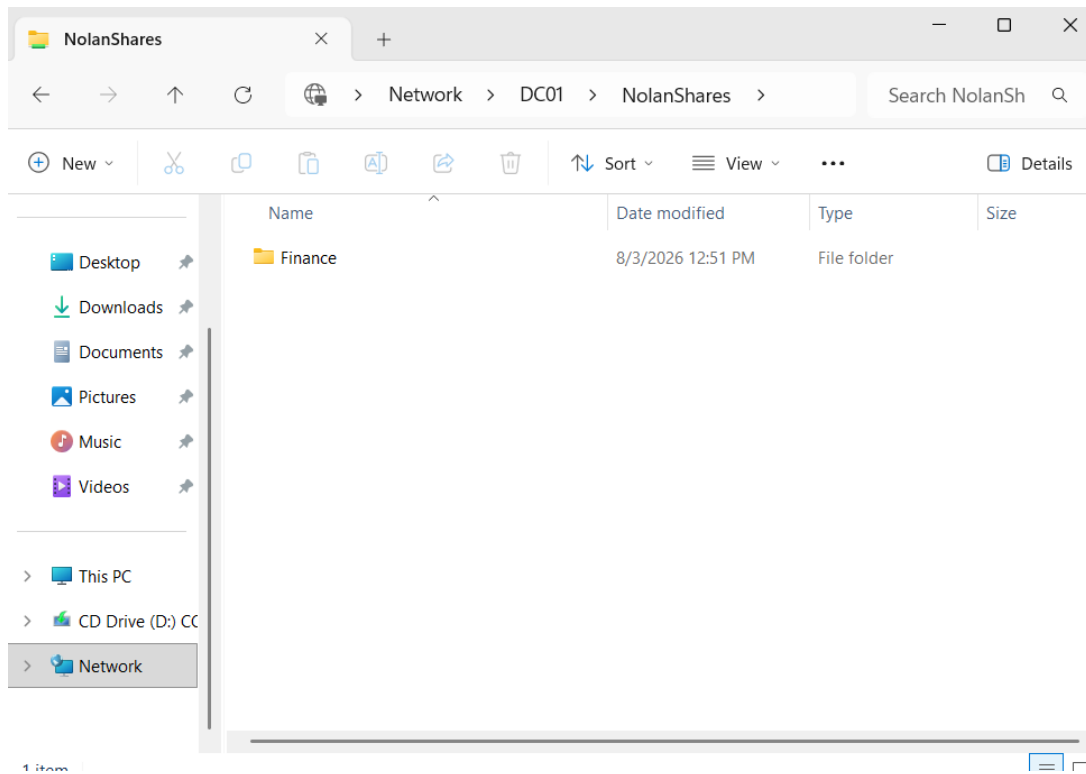


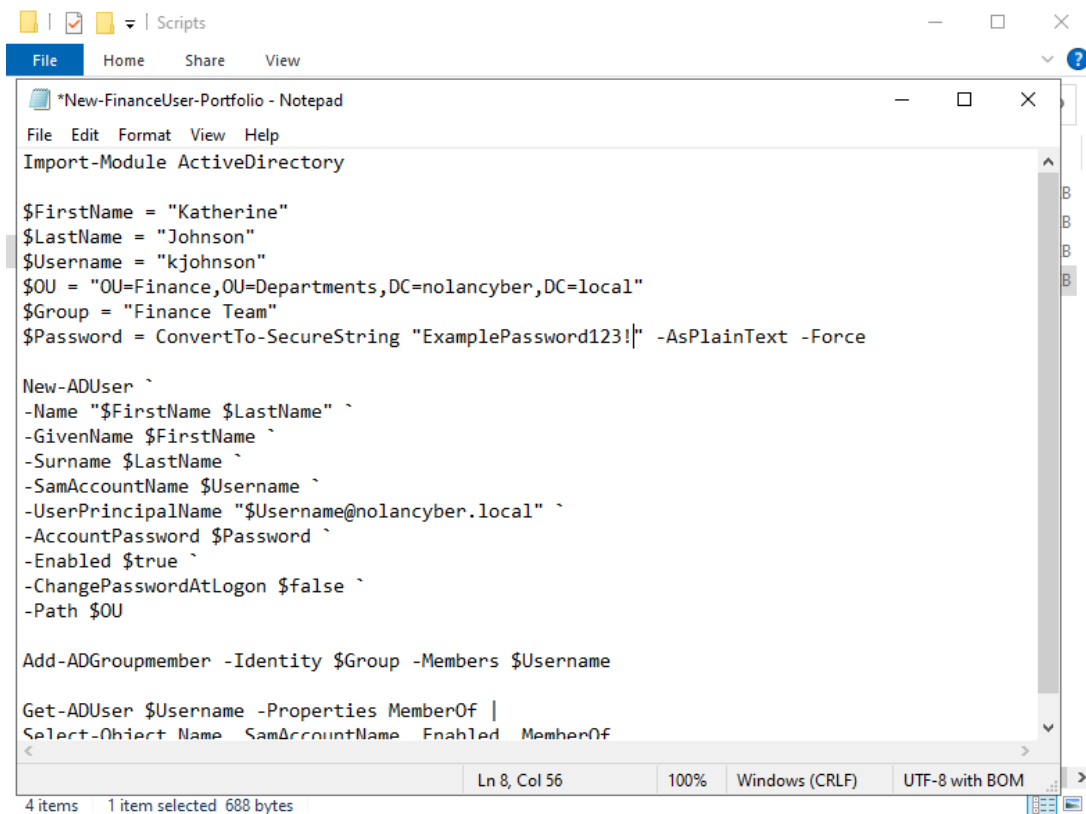
Figure 8. Departmental permission testing confirmed that the Finance user could access and create files within the assigned Finance resource.

9. PowerShell Automation

After creating users manually, I wanted to see how I could automate the process with PowerShell. I made scripts that create accounts, place them in the correct OU, and add them to the appropriate security group.

Single User Provisioning

I started with a script that creates one user so I could understand the Active Directory commands and make sure everything worked correctly. Once I got that working, I moved on to creating multiple users from a CSV file.



The screenshot shows a Notepad window titled '*New-FinanceUser-Portfolio - Notepad'. The menu bar includes File, Edit, Format, View, and Help. The script content is as follows:

```
Import-Module ActiveDirectory

$FirstName = "Katherine"
$LastName = "Johnson"
$Username = "kjohnson"
$OU = "OU=Finance,OU=Departments,DC=nolancyber,DC=local"
$Group = "Finance Team"
$Password = ConvertTo-SecureString "ExamplePassword123!" -AsPlainText -Force

New-ADUser `
-Name "$FirstName $LastName" `
-GivenName $FirstName `
-Surname $LastName `
-SamAccountName $Username `
-UserPrincipalName "$Username@nolancyber.local" `
-AccountPassword $Password `
-Enabled $true `
-ChangePasswordAtLogon $false `
-Path $OU

Add-ADGroupmember -Identity $Group -Members $Username

Get-ADUser $Username -Properties MemberOf |
Select-Object Name, SamAccountName, Enabled, MemberOf
```

The status bar at the bottom indicates 'Ln 8, Col 56', '100%', 'Windows (CRLF)', 'UTF-8 with BOM', and '4 items | 1 item selected | 688 bytes'.

Figure 9. PowerShell script used to automate creation of a single Active Directory user.

Bulk User Provisioning

After verifying the single-user script, I expanded the automation by importing employee information from a CSV file. The script automatically created multiple user accounts, placed them into the correct Organizational Units, and added each user to the appropriate security group based on department information contained in the CSV file.

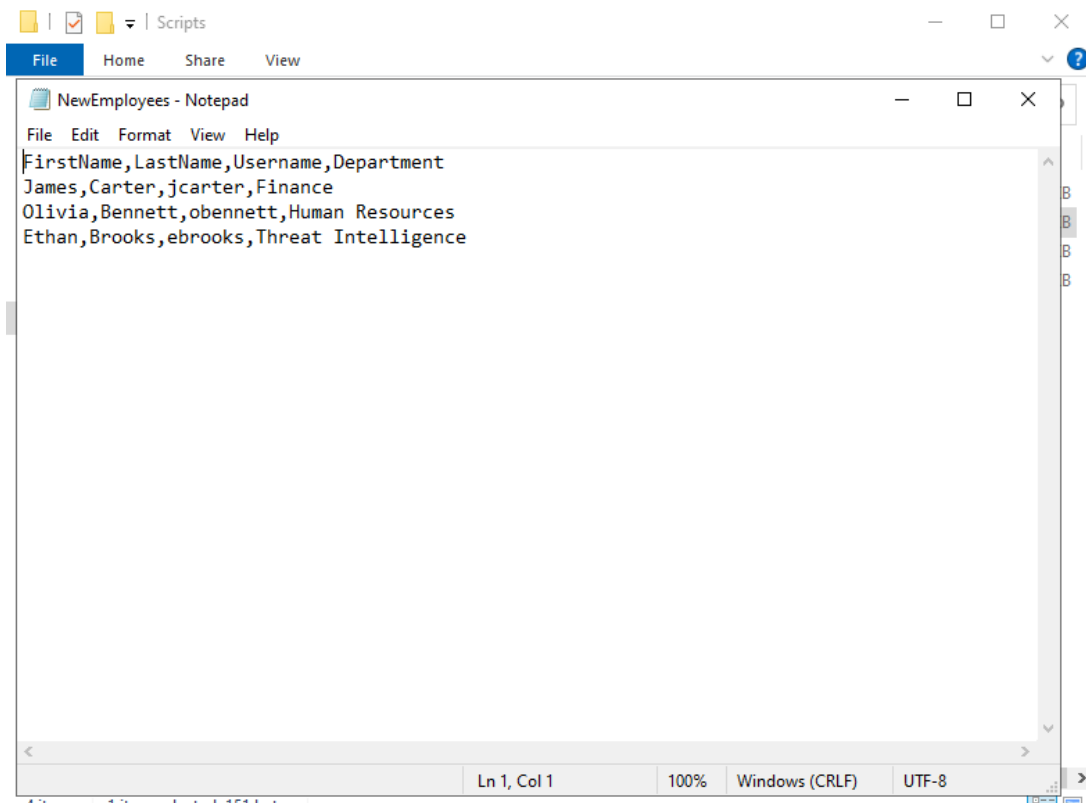


Figure 10. CSV file containing employee information used for automated account creation.

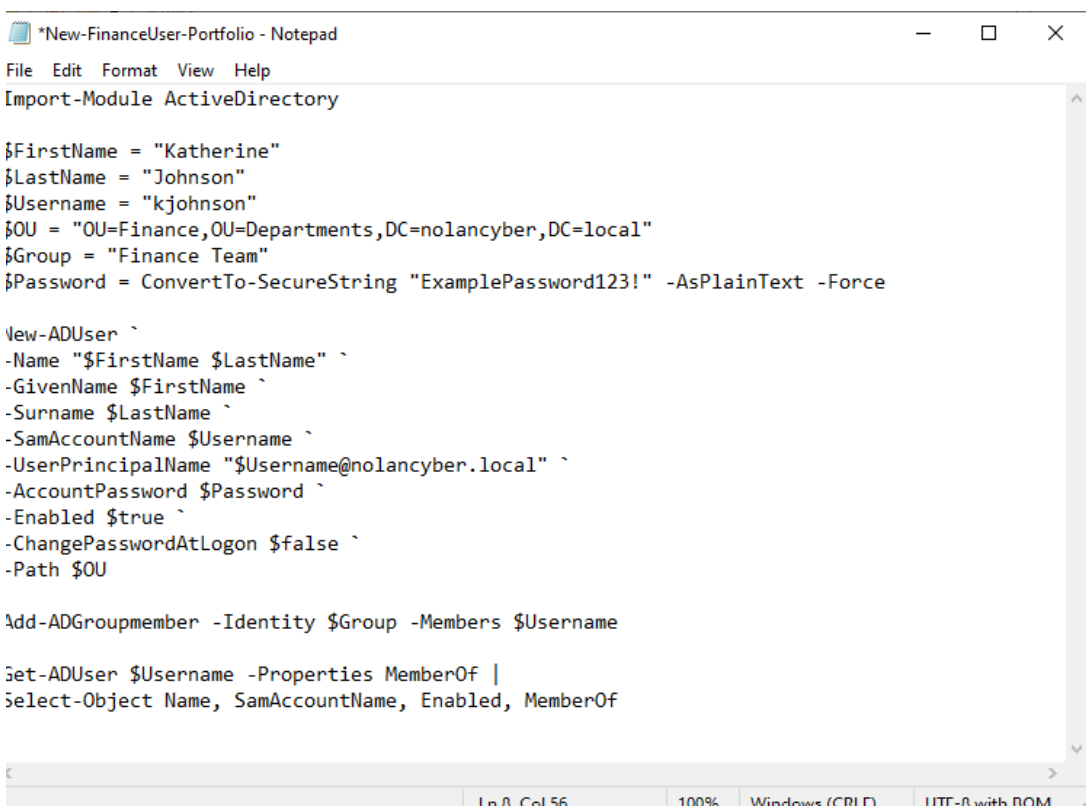


Figure 11. PowerShell script used to automate bulk user creation from a CSV file.

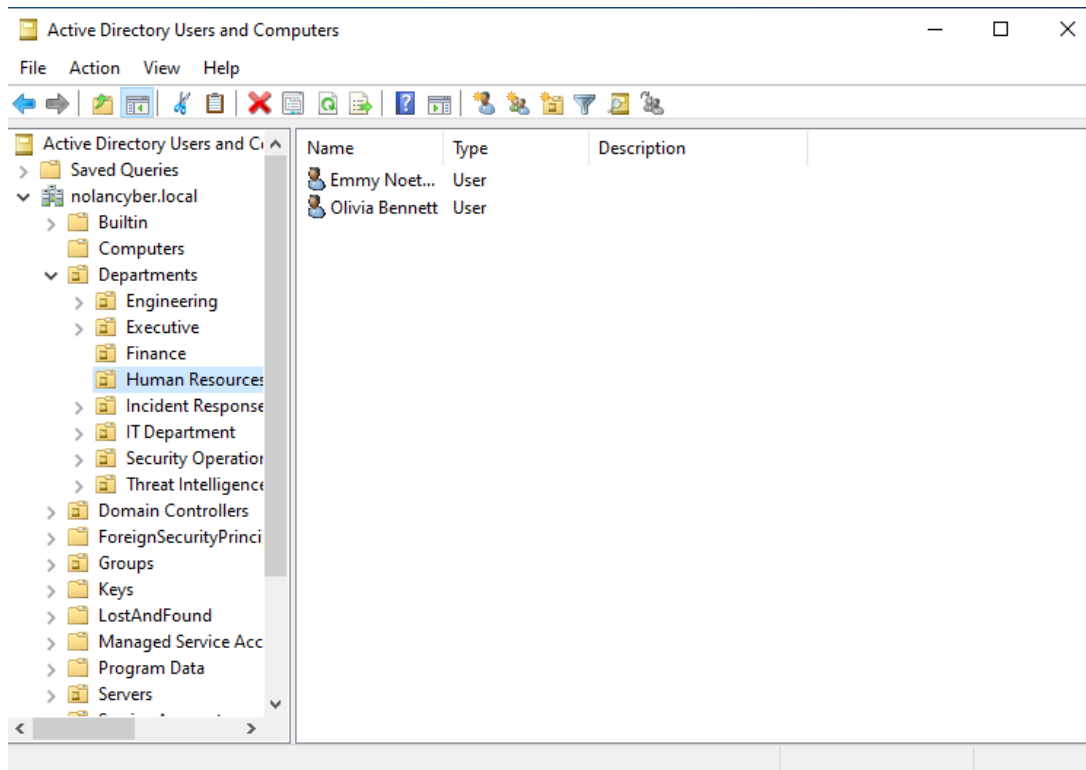


Figure 12. User accounts successfully created through PowerShell automation.

10. Security Auditing

I enabled Windows Security Auditing so I could see authentication activity in the domain. I tested successful and failed sign-ins and used Event Viewer to find the corresponding events. This also helped me learn where Windows records authentication activity and what information is available in the security logs.

Successful Logon Events

Successful logons were verified by signing in with domain user accounts and reviewing Event ID 4624 in Event Viewer. These events confirmed that authentication requests were successfully processed by the domain.

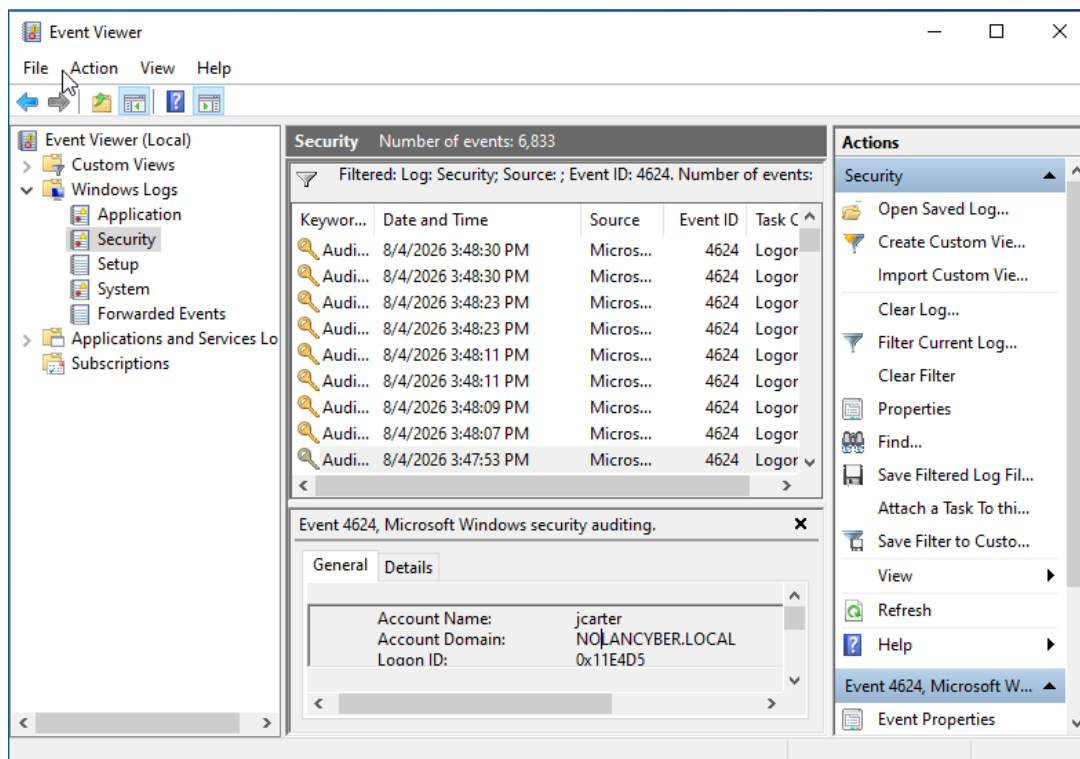


Figure 13. Event ID 4624 confirming a successful domain logon.

Failed Logon Events

Failed logons were tested by intentionally entering an incorrect password for a domain user account. Event ID 4625 was recorded in Event Viewer, confirming that failed authentication attempts were being audited correctly. This demonstrated how security logs can be used to identify unsuccessful sign-in attempts and support troubleshooting.

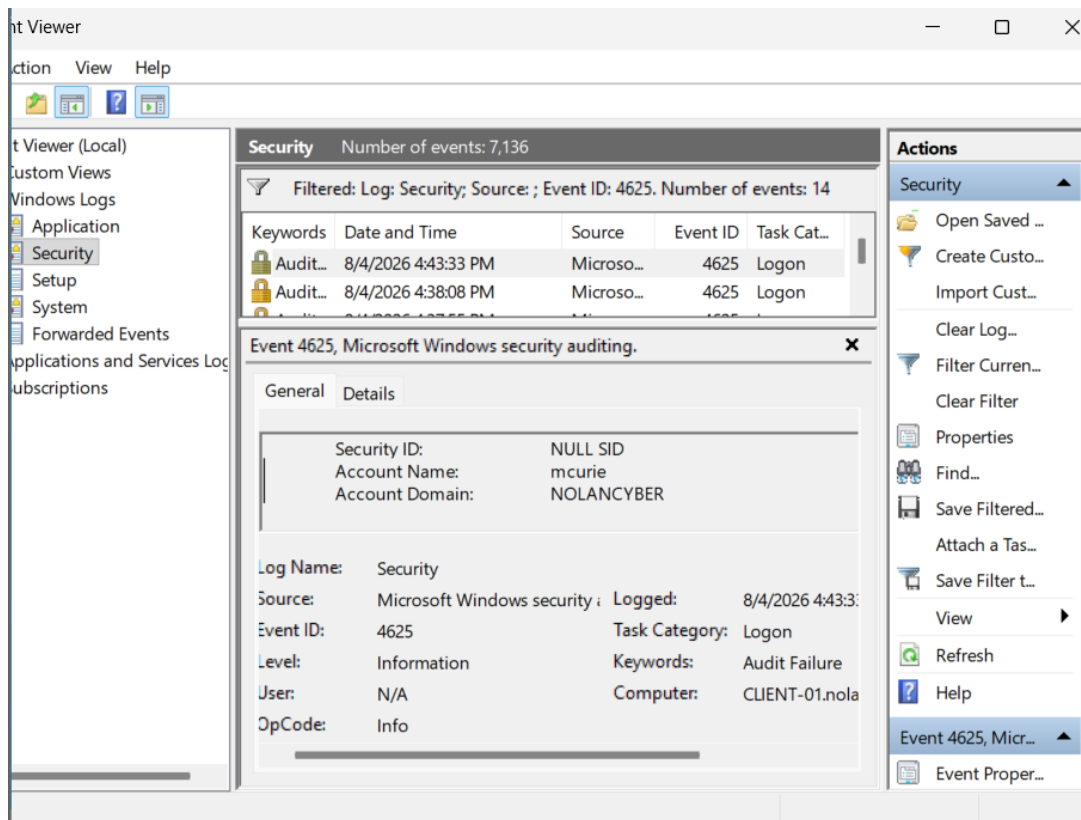


Figure 14. Event ID 4625 showing a failed authentication attempt.

11. DHCP Configuration

I installed and configured the DHCP Server role on DC01 to provide automatic IP addressing for domain workstations. I created an IPv4 scope using the address range 192.168.1.100–192.168.1.200 and configured the scope to provide DC01 (192.168.1.10) as the DNS server. After switching CLIENT-01 from a static IP address to DHCP, I verified that the client received an address from the configured scope and was using the correct DHCP and DNS servers.

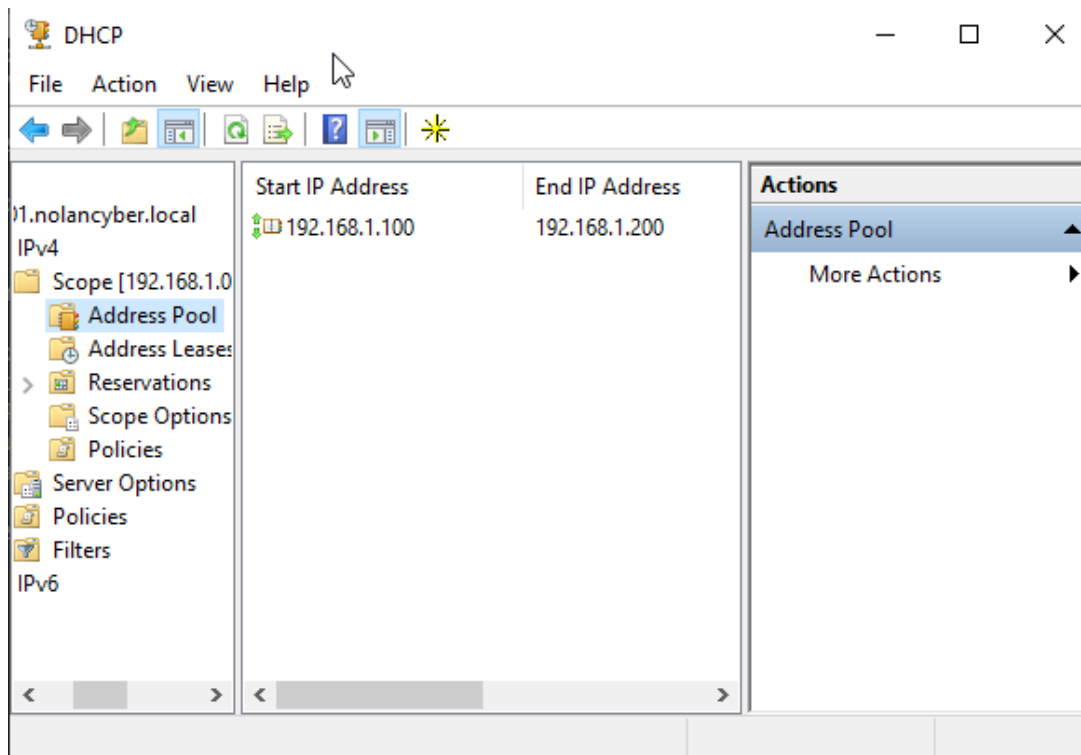


Figure 15. DHCP IPv4 scope configured on DC01 with an address pool of 192.168.1.100–192.168.1.200 for domain clients.

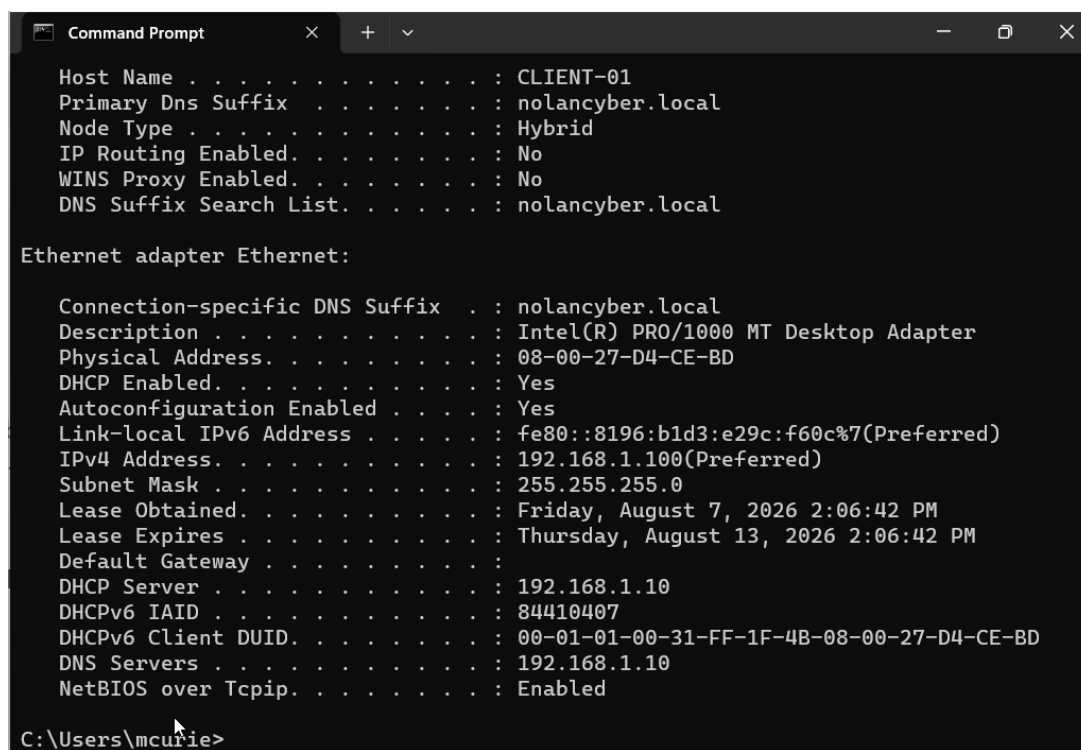


Figure 16. CLIENT-01 successfully receiving its network configuration from the DHCP server on DC01.

12. Automated Network Drive Mapping

I used Group Policy Preferences to automatically map NolanShares as the S: drive for domain users. I tested it on CLIENT-01 and confirmed that the drive appeared automatically after signing in. The existing NTFS and share permissions still controlled which department folders each user could open.

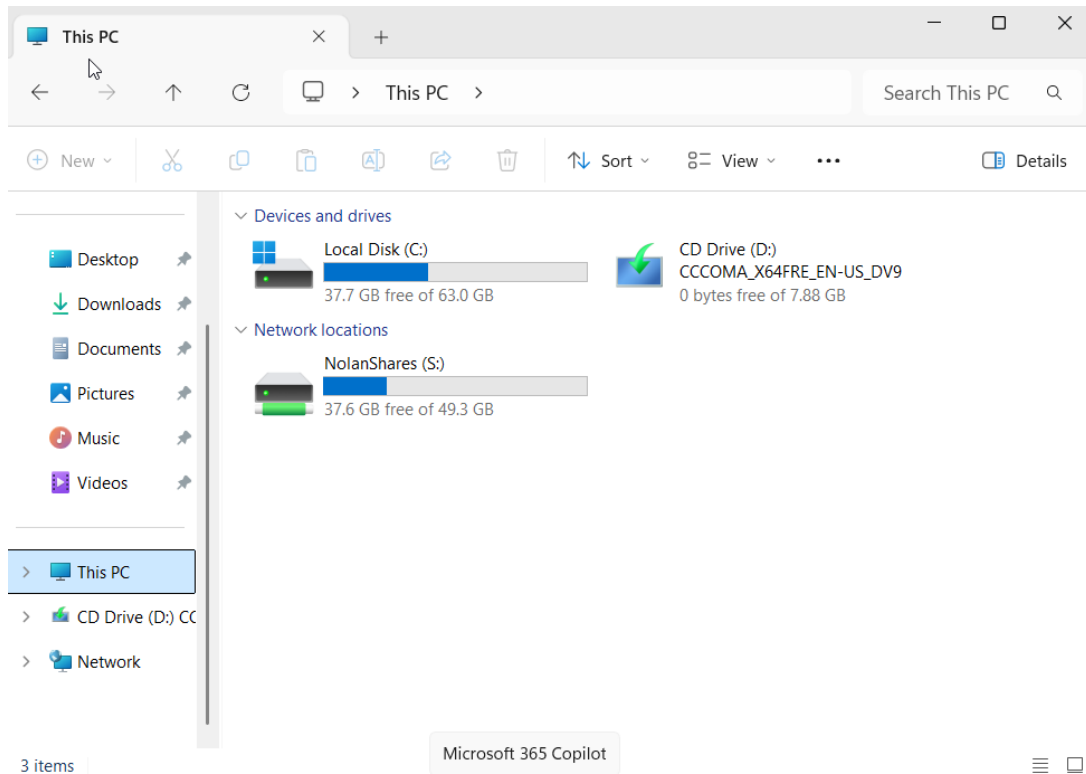


Figure 17. NolanShares automatically mapped as the S: drive on CLIENT-01 through Group Policy.

13. PowerShell Access Restriction

I created a Group Policy to restrict PowerShell access for standard departmental users while allowing members of the IT Administrators group to retain access. The policy was linked to the Departments OU, with the IT Administrators group excluded from applying the restriction. I tested the policy using both a standard Finance account and an IT account to confirm that PowerShell was blocked for the standard user but remained available to IT.

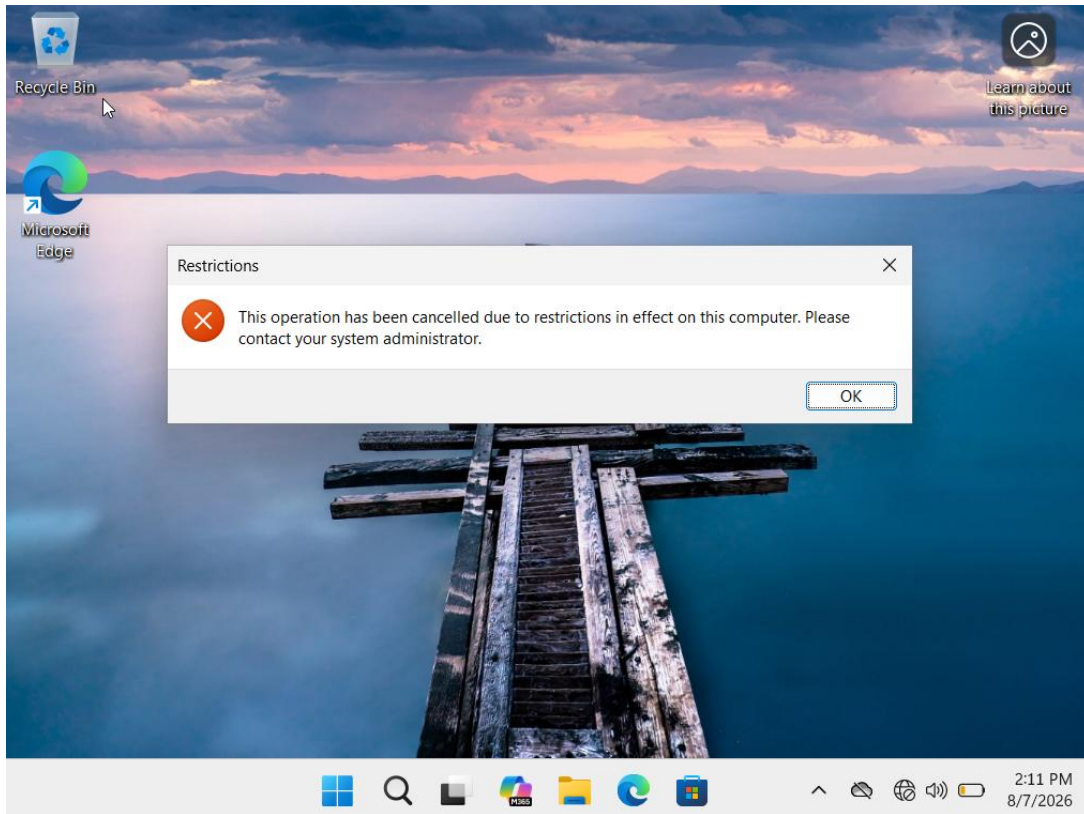


Figure 18. PowerShell access blocked for a standard Finance user through Group Policy.

14. Backup and Disaster Recovery

I installed Windows Server Backup and added a separate 25 GB virtual disk to DC01 for backup storage. I initialized and formatted the disk, then used it to create a System State backup. This gave me a recovery point for important domain controller data, including Active Directory and SYSVOL.

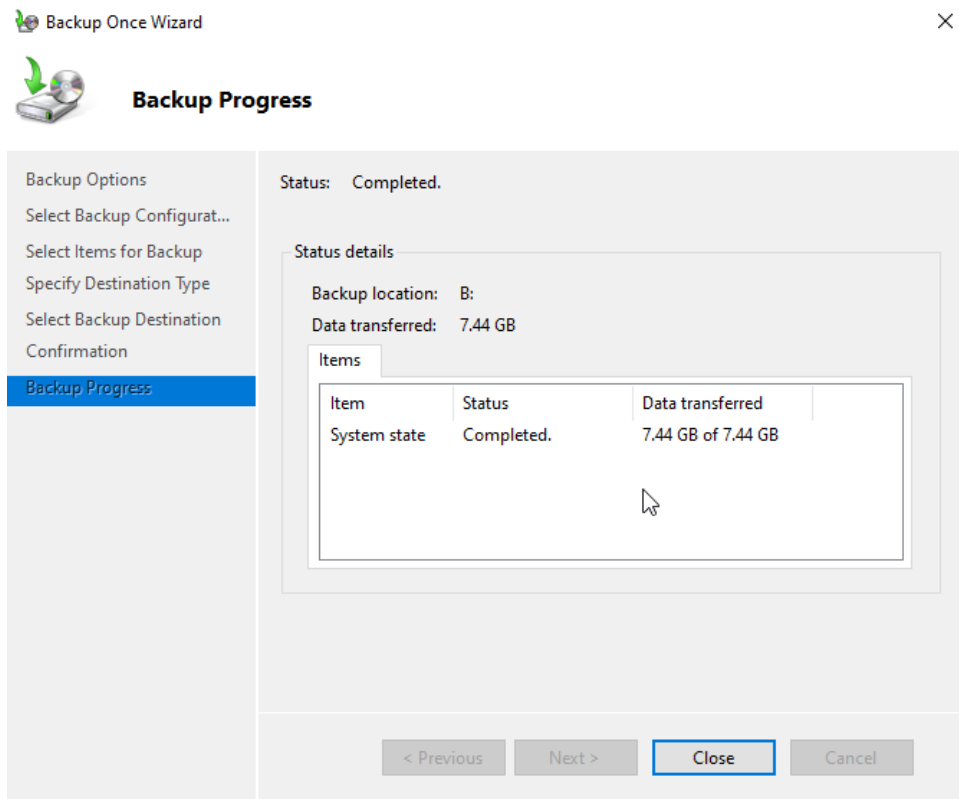


Figure 19. Successful System State backup of DC01 created with Windows Server Backup and stored on the dedicated AD Backup volume.

15. Troubleshooting

Issue	Resolution
PowerShell script errors	Corrected cmdlet parameter names and syntax.
CSV Import	Updated CSV formatting and verified OU paths and security groups.
Event Viewer	Confirmed the correct computer was being checked for audit events.
Time Synchronization	Corrected VM time configuration so both systems recorded accurate timestamps.

16. Skills Demonstrated

Core Technical Skills Demonstrated:

- Active Directory Domain Services (AD DS)
- Windows Server 2022 Administration
- Active Directory Users and Computers (ADUC)
- Group Policy Management
- User and Group Administration
- Organizational Unit (OU) Design
- DNS Configuration
- Windows Client Domain Join
- NTFS and Share Permission Management
- Role-Based Access Control (RBAC)
- PowerShell Scripting and Automation
- CSV-Based User Provisioning
- Windows Security Auditing
- Event Viewer Log Analysis
- Oracle VirtualBox Administration
- Troubleshooting and System Validation
- DHCP Configuration
- Group Policy Preferences
- Windows Server Backup

17. Conclusion

Before starting this project, I understood some of these concepts individually, but I hadn't put them together in a complete environment before. Building the domain made Active Directory much easier for me to understand because I could actually see how users, groups, permissions, policies, networking, and the client computer all affected each other.

I also learned a lot from the things that didn't work the first time. Going back through my setup and figuring out what was causing a problem ended up being one of the most useful parts of the project.